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Advanced CFD Homework #2

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November 18, 2024

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1 Introduction

This report consists of two sections regarding the performance of the DU00-W-212 airfoil in a 2D turbulent transitional flow. To study the aerodynamics of the airfoil, the simulation is first verified to produce reasonable physics at an angle of attack $\alpha = 0^\circ$, before using numerical results from [1] to validate the simulation at an angle of attack $\alpha = 4^\circ$. Once validated, the second part of the report focuses on optimizing the lift-to-drag ratio (C_l/C_d) by modifying the airfoil geometry. The adjoint optimization process is utilized for this purpose, and its accuracy, as well as reliability, are investigated before summarizing the report with an improved airfoil design.

STAR CCM+ is the CFD software used in this report, along with [2] and [3] documenting the development of the simulation.

2 Part 1 - Transition

The 2D airfoil transition simulation employs the SST (Menter) $k - \omega$ model for turbulence and the Gamma-ReTheta model for transition. These models are essential for accurately capturing the flow behaviour. The results of this simulation offer insights into the airfoil's aerodynamics, which are explored in the following section.

2.1 Post-Processing of Turbulence Modelling and Transition

As the airfoil is subject to the flow, a boundary layer forms around its surface. This thin layer of fluid, impacts both the drag and lift forces acting on the airfoil. Further from the airfoil, the fluid behaves as shown in the velocity and pressure distributions in Figure 1. These plots provide insights into key flow features like stagnation points and separation points.

A stagnation point is a point in the flow field where the fluid's velocity is zero. Considering the velocity distribution in Figure 1a, the stagnation point is at the LE ($x = 0$); as illustrated in Figure 2a the velocity is zero in magnitude and hence stagnant.

Conversely, separation points are defined as locations along the surface where the boundary layer of a fluid flow detaches. Detachment occurs when the pressure gradient is adverse, meaning the flow reverses direction. This occurs at the airfoil's TE, as visualized by the vectors pointing in the opposite direction of the flow in Figure 2b.

Another method for understanding the aerodynamic performance of an airfoil is to plot the pressure and friction coefficients, C_p and C_f respectively, along the airfoil's surface x .

The pressure distribution in Figure 3a provides several insights; the location of maximum C_p , which is at the LE, indicates the stagnation point. Also, the region where the pressure increases, implies an adverse pressure gradient which can lead to the separation point. Moreover, the difference in pressure from the upper to the lower surface is the amount of lift created by the airfoil.

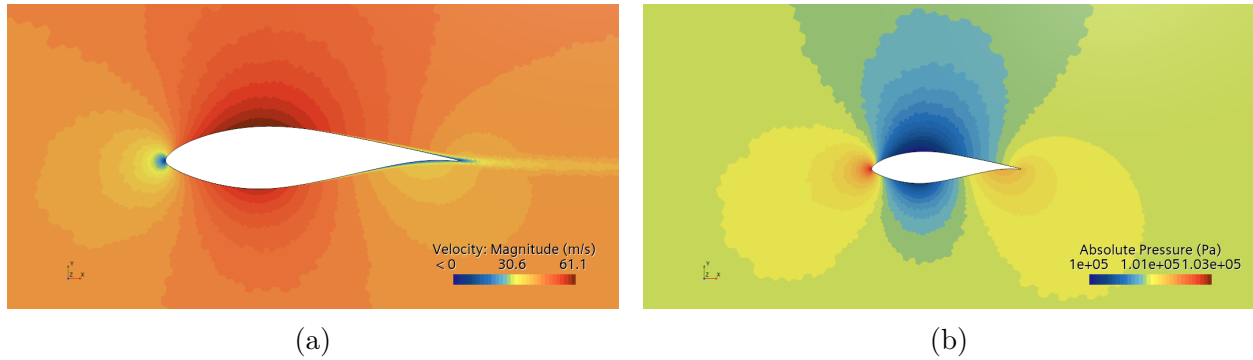


Figure 1: Scalar plots illustrating the distribution (a) Velocity magnitude [m/s] and (b) Pressure [Pa] in the flow around the airfoil.

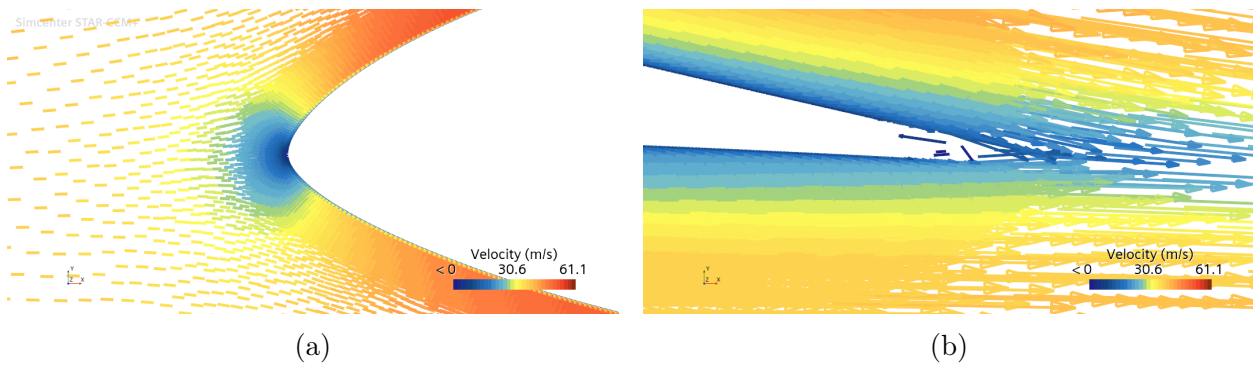


Figure 2: Velocity vector scenes to display the flow features (a) stagnation point at LE and (b) separation points at TE.

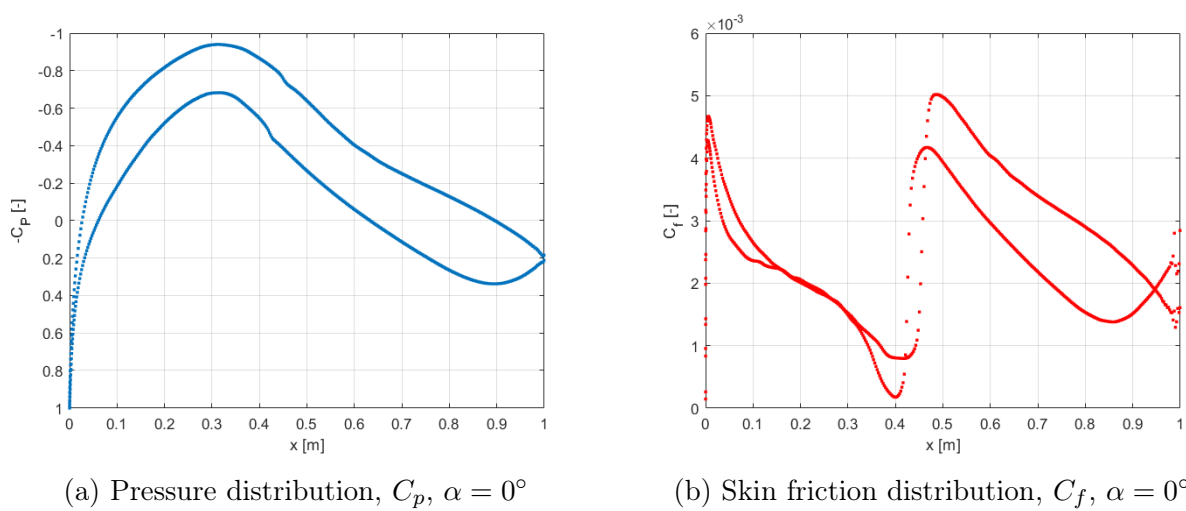


Figure 3: Pressure and skin friction distributions on the airfoil.

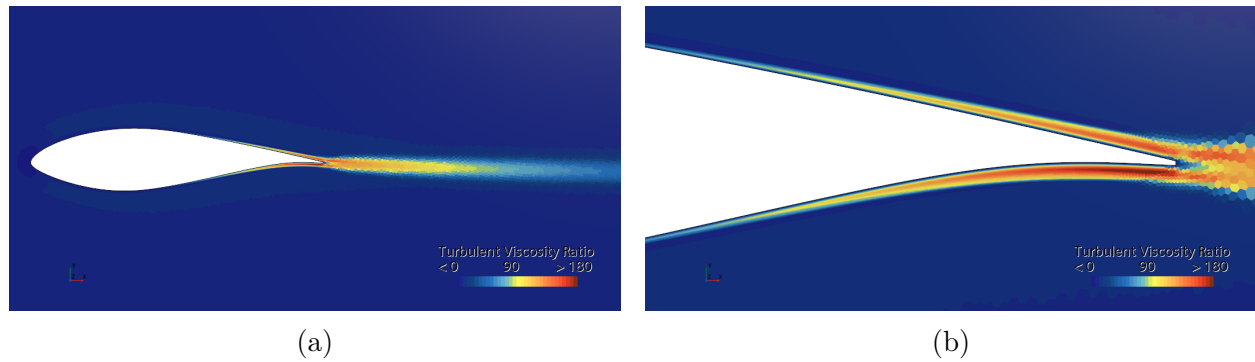


Figure 4: Scalar plot of Turbulent Viscosity Ratio ($\frac{\nu_t}{\nu}$) for (a) flow around the complete airfoil and (b) flow near TE.

On the other hand, the friction distribution from Figure 3b provides insight into skin friction drag contributions. The amount of friction indicates the flow type, where a larger friction coefficient indicates turbulence.

The fluid flow around the airfoil is characterized by two distinct types, laminar and turbulent, which depict how fluid particles move and mix. Turbulence occurs at high Reynold numbers (ratio of inertial forces to viscous forces) and has an inherently complex and chaotic nature that cannot be solved analytically. Instead, a model is needed to close the problem and resolve the flow. The previously mentioned model is from the eddy viscosity class of RANS. These models represent turbulence by introducing a hypothetical property of the fluid, known as turbulence viscosity ν_t .

One method of visualizing areas of turbulent flow is a scalar plot of the turbulent viscosity ratio ($\frac{\nu_t}{\nu}$). A high ratio indicates that turbulent effects dominate over molecular diffusion and the flow is turbulent; vice versa a low ratio indicates laminar flow. Consider Figure 4, showing that turbulence is mostly present aft of the airfoil and the TE.

Flow that features both laminar and turbulent flow, contains a transition region where the flow is a mixture of both flow types. A method of visualizing this region is intermittence which describes the flow type. Turbulent flow is 1 and 0 is laminar. Consider Figure 5a indicating a laminar boundary layer near the LE of the airfoil. The boundary layer is thick since viscous forces are dominant close to the surface and only at a significant distance from the airfoil, the flow turns turbulent. On the other hand, a turbulent boundary layer, present aft of the airfoil, shown in Figure 5b is significantly thinner since here inertial forces dominate.

Based on the intermittence plot, the transition region is seen in Figure 5c to begin at roughly $x = 0.25c$ as spots with values between 0.1 and 0.5 are seen to form. This can be seen in Figure 5d at the end of the transition region at roughly $x = 0.44c$ where the intermittence approaches a value of 1. Returning to the skin friction coefficient plot in Figure 3b, similar x values can be seen for the start and end points of the transition, as the friction rapidly changes.

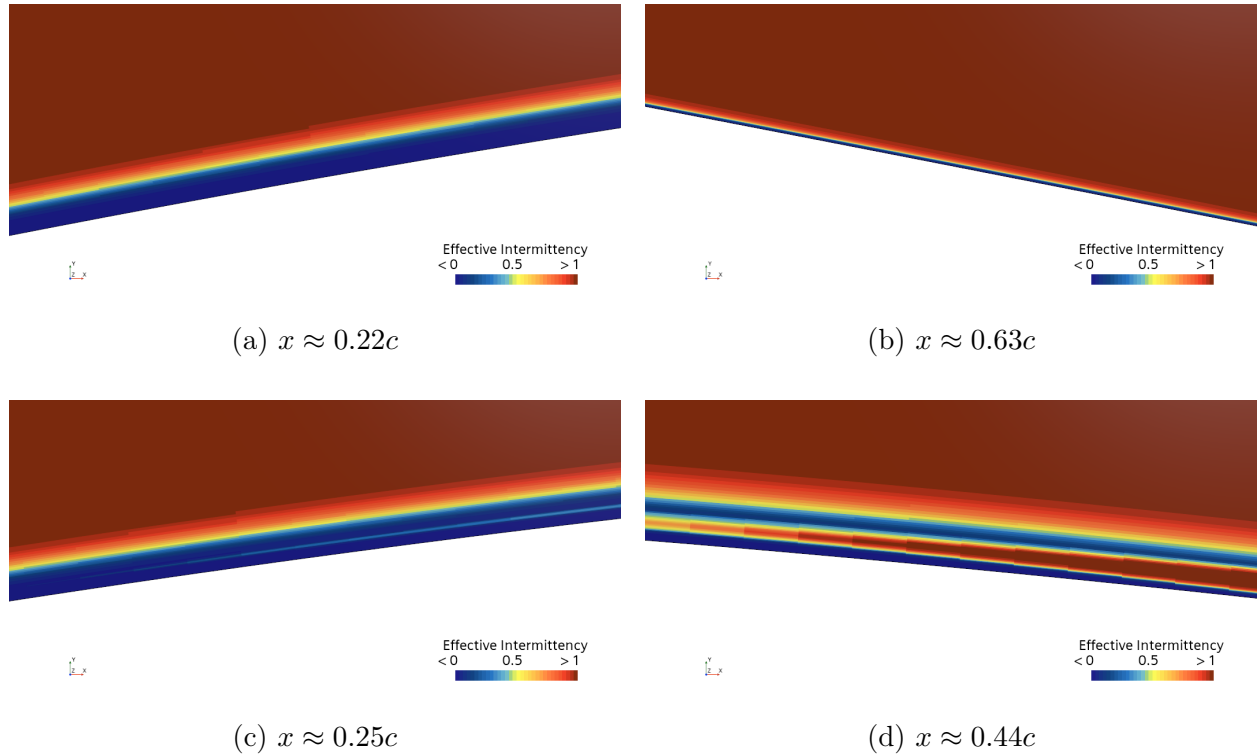


Figure 5: Scalar plot of intermittence near the airfoil surface x showcasing (a) fully laminar flow region, (b) fully turbulent flow, (c) start of transition region and (d) end of transition region

2.2 Verification of Simulation ($\alpha = 0^\circ$)

2.2.1 Mesh convergence study

Studying the convergence provides insight into whether the solution is independent of the mesh size and adequately represents the physics. The number of iterations needed to obtain the solution can be defined by certain tolerance criteria. Here a tolerance of 1×10^{-3} for all residuals has been chosen. As shown in Figure 6 the coefficients C_L and C_D have reached a plateau before reaching the convergence criteria. This took almost 2500 iterations.

To verify if the solution is independent of the mesh, the simulation ran for cell sizes of 0.5 [m] and 1 [m] while keeping the thickness constant at 10 [mm]. Consider Figure 7 showing that the drag coefficient C_d converged to the same solution while the lift coefficient differs by a negligible 0.0042. Comparing both coefficients with values provided in [1], equivalent results are noticed.

In turbulence, non-dimensional variables such as y^+ provide a framework for analyzing the flow, specifically wall-bounded flows such as airfoils. Wall-bounded flows feature the viscous sublayer where an expression given by $U^+ = y^+$ is said to accurately model the flow up to approximately $y^+ = 1$. To obey this expression, turbulence models like the one used here, require the first cell's centroid to be located at a height lower than 1. This is seen to hold along the airfoil's surface in Figure 8. It is interesting to note that the shape strongly

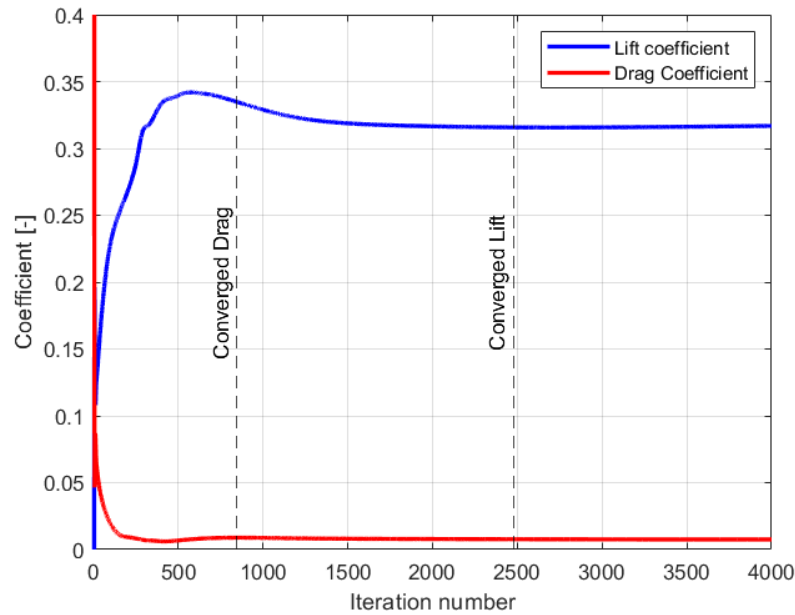


Figure 6: Aerodynamic coefficients lift C_l and drag C_d with number of iterations. The convergence line is shown for a tolerance of 1×10^{-3} .

resembles the friction coefficient distribution from Figure 3b because y^+ contains the term τ_w from the definition of friction coefficient.

2.2.2 Sensitivity Analysis

Another way to verify the simulation is to assess whether the solution results are reasonable. Altering the turbulence intensity from 0.5% to 5% and running for simulation without the transition model yield the results shown in Table 1.

Intensity	0.5%	5%	5% & Fully turbulent
C_d	0.007512	0.0078846	0.01196

Table 1: Drag coefficients C_d for different turbulence intensities.

For an airfoil at zero angle of attack, predicting how C_d will change with greater turbulence intensity is rather complex. More turbulence leads to less pressure drag by delaying the separation point further down the aft of the airfoil [4]. However, skin friction drag is dependent on the wall shear stress, which increases with turbulence intensity. For a streamlined body such as an airfoil, it can be assumed that skin friction drag has a larger significance than pressure drag and therefore C_d will increase. This is supported by the finding shown in Table 1, however, it cannot be guaranteed. On the other hand, excluding the transition model is likely to increase C_d because the separation point will occur earlier, leading to more pressure drag.

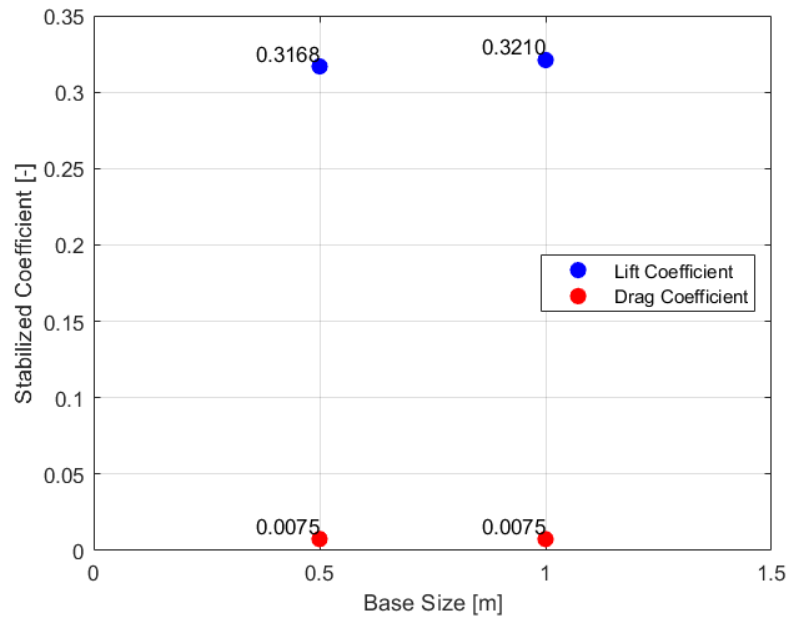


Figure 7: Converged aerodynamic coefficients lift C_l and drag C_d as a function of the number of cells for the fixed thickness of prism layers

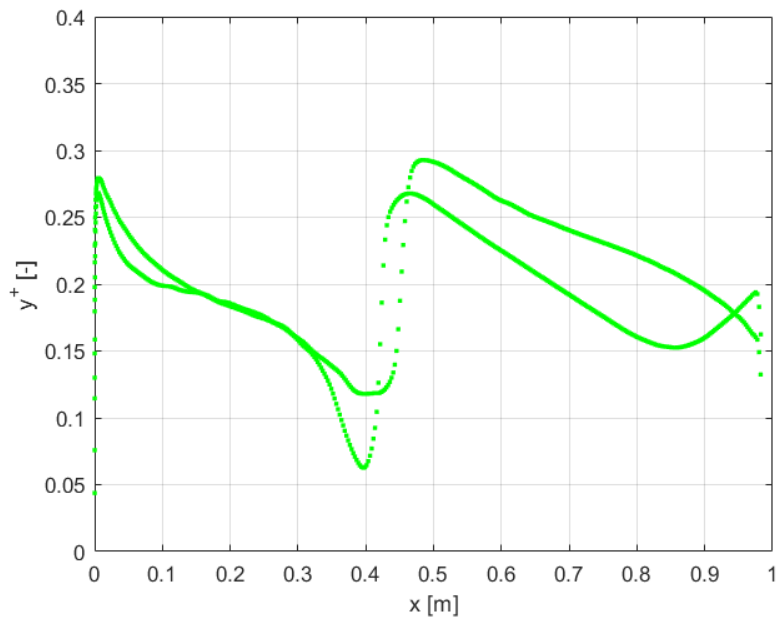


Figure 8: Dimensionless wall normal $y^+ = \frac{y_p U_\tau}{\nu}$ where $U_\tau = \sqrt{\frac{\tau_w}{\rho}}$ is the friction velocity and y_p is the distance to the first cell centroid.

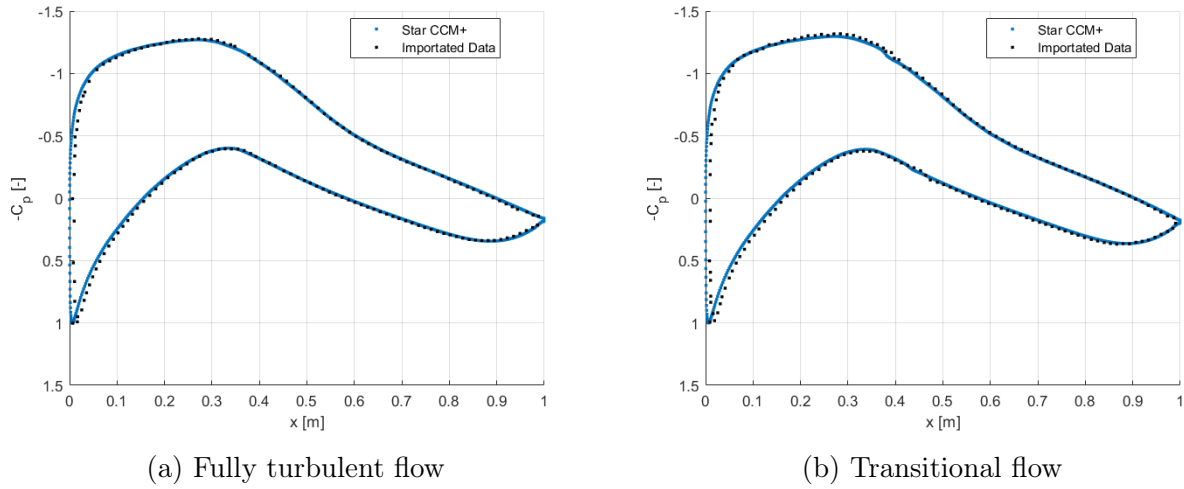


Figure 9: Pressure distributions C_p , $\alpha = 4^\circ$ on the airfoil from Star CCM+ compared with imported data from [1].

2.3 Validation transition modeling ($\alpha = 4^\circ$)

Now that the flow around the airfoil for zero angle of attack has been analyzed and verified to yield physically reasonable results, the simulation can be validated using data from [1]. For a different point of comparison, the angle of attack has been changed to 4° by adjusting the input velocity. In the same manner, as for zero angle of attack, the aerodynamic coefficients shown in Table 2 are obtained for fully turbulent flow and transitional flow.

As previously explained, the drag is expected to decrease for the transitional flow. On the other hand, the lift coefficient C_l increases for the transitional flow. Both of which are seen in the results. In transitional flow, the adverse pressure gradient occurs further down the aft of the airfoil leading to delayed flow separation and hence more lift [4].

	Fully Turbulent	Transitional
C_d	0.01465	0.00942
C_l	0.78578	0.79380

Table 2: Comparison of aerodynamic performance for fully turbulent and transitional flows at $\alpha = 4^\circ$.

Comparing these values with data from Figure 4.5 and 4.7 in [1], the similarity in the values is noticeable.

For further validation with [1], the pressure distribution is plotted for fully turbulent and transitional flow in Figure 9. A strong correlation with almost perfect overlap can be seen between the two, once again validating the simulation.

3 Part 2 - Adjoint Optimization

The adjoint method is a highly efficient optimization technique for predicting the influence of design variables and physical inputs on the objective of the simulation. In other words, it predicts the sensitivity of the output with respect to the input variables, offering valuable insights into the performance. Unlike traditional optimization techniques that commonly require multiple simulations for each design variable, the adjoint method requires only a fully converged flow solution and a single adjoint solution. This makes it computationally cost-effective for evaluating the sensitivity of output even when increasing the number of design variables [5]. In this part of the report, the simulation aims to maximize the lift-to-drag ratio by modifying the geometry of the DU00-W-212 airfoil.

3.1 Results & Reliability of Adjoint Optimization

The primary flow simulation described in the previous section, took a total of 4000 iterations to solve the governing equations. Based on this solution, the lift-to-drag ratio function is defined for the adjoint simulation. The adjoint equations were then solved in approximately 110 iterations until convergence was reached. Consider Figure 10, which shows the residuals for both the primary and adjoint solutions, with the latter displayed in the zoomed-in box.

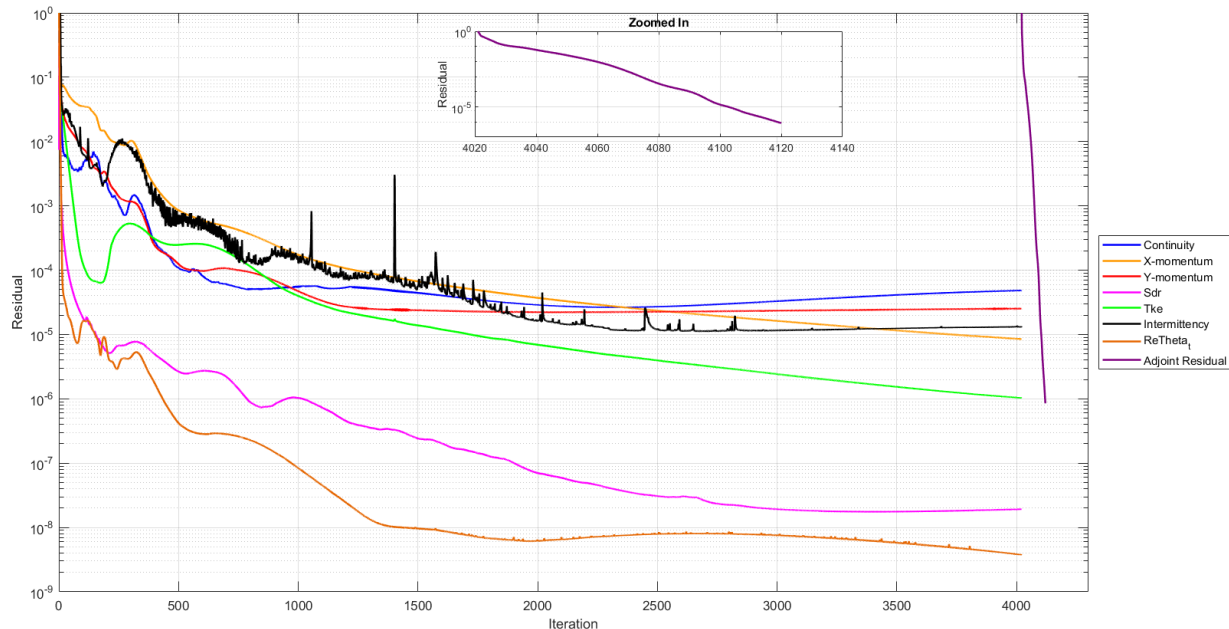


Figure 10: Residuals for the primary and adjoint solution

The residuals from the primary solution show a typical convergence behaviour, as they rapidly decrease in the initial stages of the simulation and stabilize at lower values as the

simulation runs. Continuity and X,Y momentum residuals stabilize around 1×10^{-4} , while the Turbulence residuals drop below 1×10^{-7} . These residual values are considered an acceptable level of accuracy for this paper. Moreover, the adjoint solution, which specifically tracks the optimization of the lift-to-drag ratio as a function of the changes in the airfoil geometry, reached a satisfactory tolerance of 1×10^{-6} , ensuring the reliability of the adjoint optimization process.

3.2 Suggested improvements

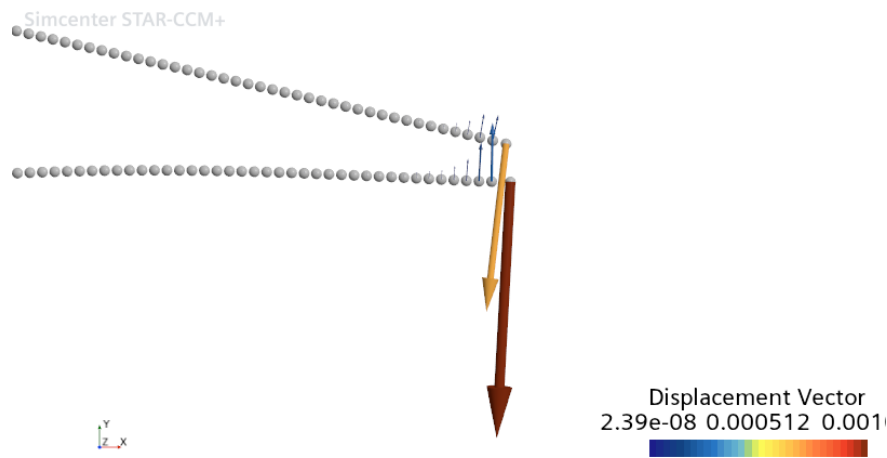


Figure 11: Adjoint Solution change suggestion on the airfoil Surface Points

The adjoint equations, solved after defining the design parameters, provided the gradients/sensitivity of the mesh. These indicate which points of the geometry have the most significant contribution to the lift-to-drag ratio. Based on the size of the contributions, adjustments to the airfoil geometry are recommended by deforming the mesh with the specified morphing scales. Four mesh adjustments were made based on a morphing scale of 2×10^{-7} and one for 5×10^{-7} , yielding a total of five iterative steps.

The shape changes are likely attributed to the importance of the TE on both the wake formation and the drag generated on the back side of the airfoil. More importantly, the TE has a great influence on the lift by redistributing the pressure more favourably. According to the point set vector scene in Figure 11, the displacement vectors on the lower surface of the TE have a larger magnitude compared to the ones on the upper surface. These displacement vectors suggest that a downward extrusion of the TE is the recommended shape modification to optimize the performance of the airfoil.

3.3 Optimization Analysis

The optimization aims to maximize the lift-to-drag ratio $\left(\frac{C_l}{C_d}\right)$ after each iteration step shown in Figure 12a. The ratio increases from 43.01 to 68.2, which is a significant increase of

58%. To determine what contributes to the increase in the lift-to-drag ratio, the aerodynamic coefficients are plotted individually as shown in Figure 12b.

Both coefficients increase with the number of iterations. In an ideal case, the lift coefficient would increase and the drag coefficient decrease. However, taking into account the order of magnitude of the two coefficients, increasing the lift coefficient at the cost of an increase in drag still yields a more optimal lift-to-drag ratio. Overall the lift coefficient increases from 0.320 to 0.584, while the drag increases from 0.0075 to 0.0086.

For further investigation into the cause of a more optimal lift-to-drag ratio, the pressure and friction distributions for all five optimization steps are plotted in Figure 13. The pressure distribution in Figure 13a shows that the pressure on the upper side of the airfoil becomes lower with each optimization step, while the lower side's pressure increases. These changes cause a larger pressure imbalance along the surface of the airfoil, causing more lift. On the other hand, the skin friction distribution in Figure 13b behaves roughly the same until the transition point. After this, the skin friction is seen to significantly increase on the upper surface and decrease on the lower surface with each optimization step. Nevertheless, the total friction increases suggesting that the change on the upper surface is more significant than on the lower.

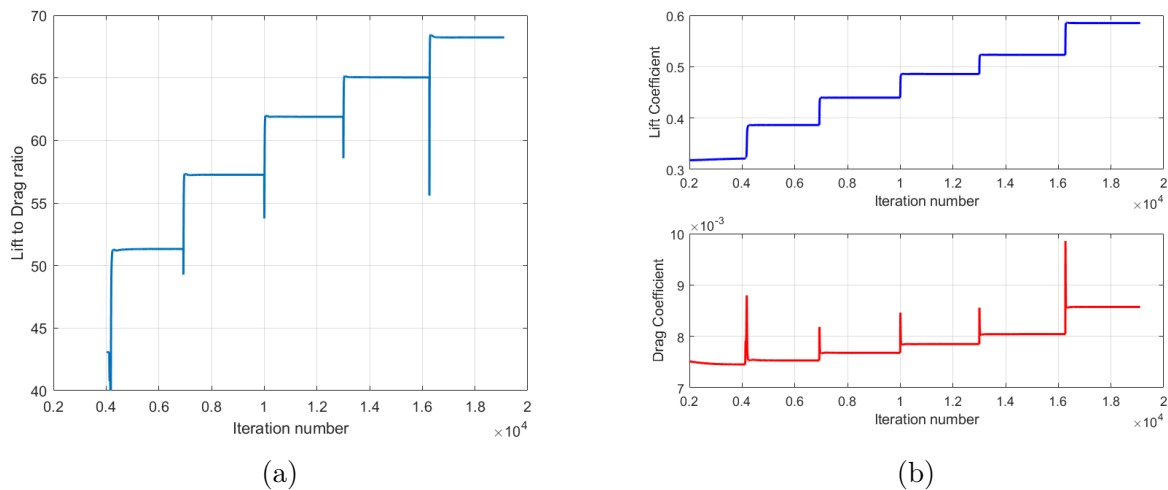


Figure 12: Changes in aerodynamic performance (a) C_l/C_d ratio, (b) C_l and C_d coefficients across iteration numbers for five adjoint optimization steps.

The proposed airfoil modifications strongly resemble the shape of a Gurney Flap, which is a small perpendicular segment attached to the TE of an airfoil. Commonly used in the aerospace and automotive industries, Gurney Flaps are known to improve the performance of aircraft wings and car spoilers, popularized by race car driver Dan Gurney [6]. A visual comparison is provided in Figure 15.

This geometry significantly enhances the maximum lift and improves the aerodynamic efficiency by creating a larger pressure differential between the upper and lower surfaces of the airfoil. It also stabilizes the airflow around the TE, reducing drag and delaying the onset of flow separation [6]. According to Wang, the maximum lift coefficient of a regular

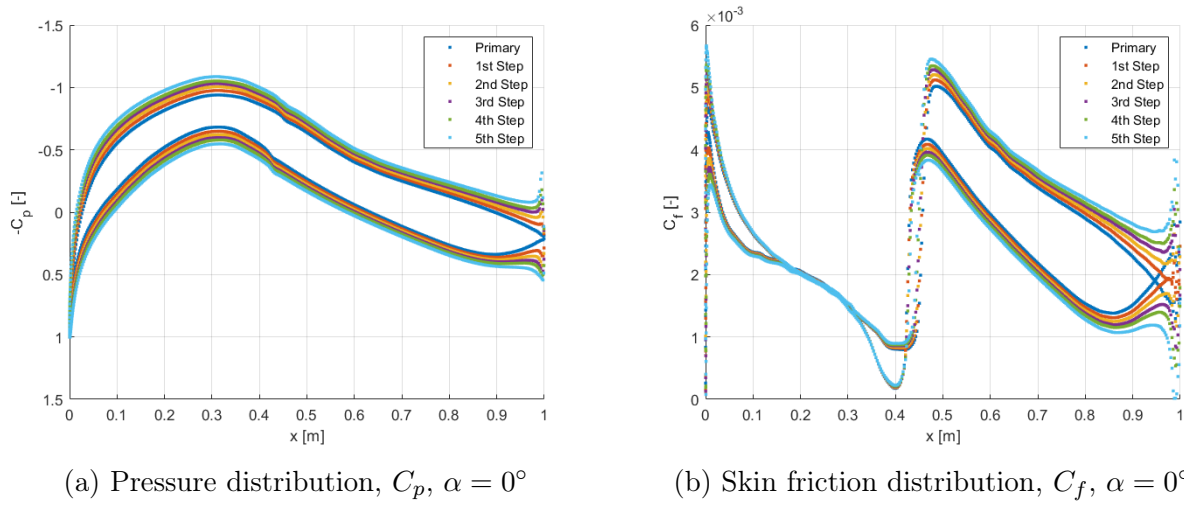


Figure 13: Pressure and skin friction distributions on the airfoil for primary and adjoint optimization.

NACA0012 airfoil at $\alpha = 0^\circ$ fitted with a Gurney flap increased by 10% [6]. Considering the original airfoil had a blunt trailing edge, the optimization resembling a Gurney flap led to a 82.23% lift coefficient total increase, with a 20.39%, 13.78%, 10.49%, 7.68%, and 11.82% increase in every iteration respectively.

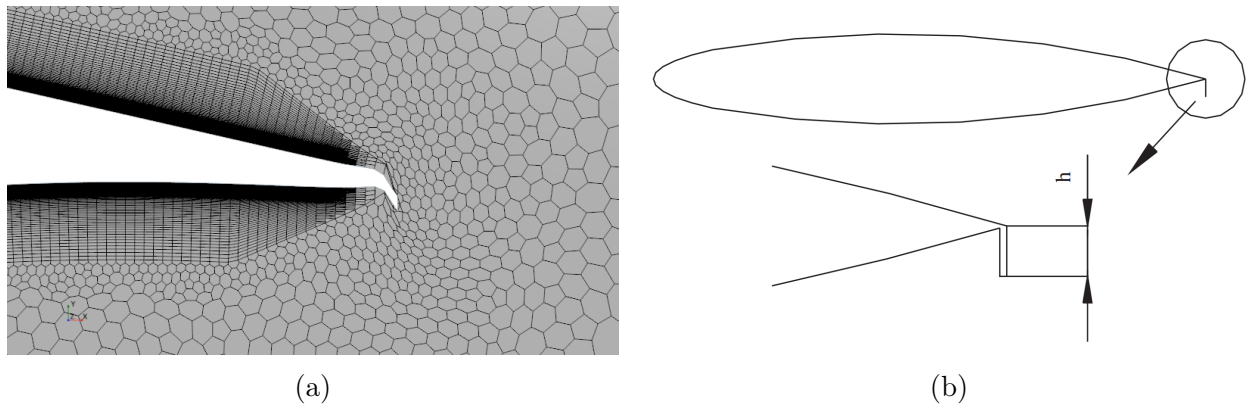


Figure 14: TE obtained by (a) adjoint optimization simulations and (b) Gurney Flap study [6]

3.4 Mesh Refinement

The adjoint optimization method is limited by its dependence on the primary solution of the flow. This means that any inaccuracies in the mesh of the primary solution will impact the result of the adjoint cost functions that work on optimizing the lift-to-drag ratio. To refine the mesh and limit the impact of numerical inaccuracies, the adjoint error estimate,

shown in Figure 15b, and sensitivity, shown in Figure 15a, indicate where the mesh would benefit from mesh refinement.

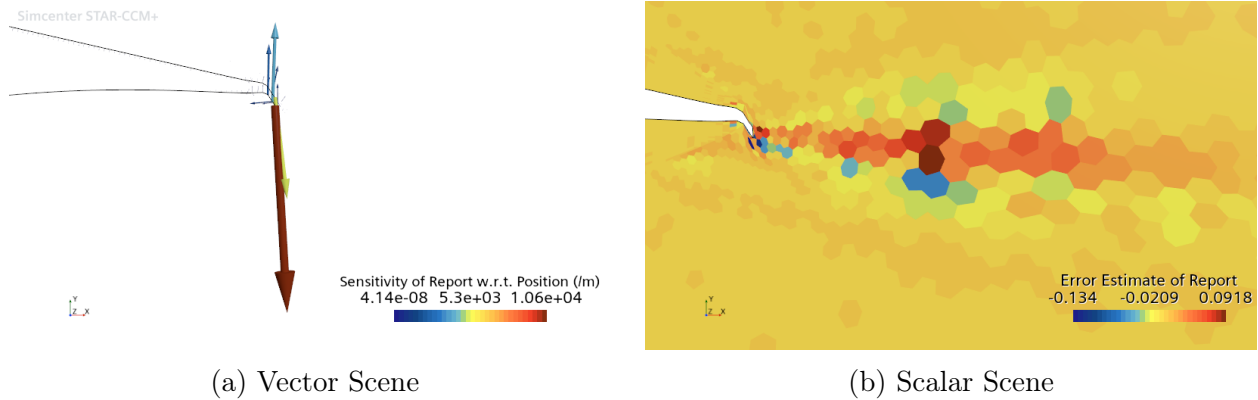


Figure 15: Scenes illustrating possible error sources and areas of improvement through (a) Sensitivity to position and (b) Error estimate.

3.5 Automation

The entire process of optimization shown in this report, including mesh refinement, may be optimized using HPC in the following procedure:

- **Primary Solver:** STAR-CCM+ is automated to set up the geometry and mesh and solve the primary flow model in parallel. This means multiple processor cores are used simultaneously.
- **Adjoint Solver:** HPC queuing systems such as LSF can be automated to run the adjoint solver based on the output of the primary solver, without manual intervention.
- **Post-Processing:** Extraction of important results from both solvers can also be automated using batch scripts or commands. The recommendations for the mesh are then applied, along with potential refinements.
- **Iterative optimization:** Queuing systems are once again used to repeat this process until the design specifications are met.

4 Conclusion

Results obtained by SST (Menter) $k - \omega$ and Gamma-ReTheta model have proven to be physically reasonable and show strong agreement with numerical results in [1]. This consistency has been enough to validate the reliability and further use of the solution in adjoint optimization. The adjoint simulation successfully optimized the existing DU00-W-212 airfoil geometry through mesh deformation, to maximize the lift-to-drag ratio. The final optimized geometry features a downward extrusion at the TE, a modification that aligns with the principles of airfoil improvement provided by the Gurney flap [6].

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Nomenclature

CFD Computational fluid dynamics

HPC High Process Computing

LE Leading Edge

LSF Load Sharing Facility

TE Trailing Edge

References

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